MAILTRACE AI // FORENSIC TELEMETRY PLATFORM

Digital Evidence & Forensic Incident Dossier

REPORT REFERENCE:	RPT-■20260917-■B31D9B	**GENERATED (UTC):**	2026-■09-■17 18:■31:■07 UTC
EVIDENCE IDENTIFIER:	EVD-■5E58C282C0	**ASSIGNED CASE:**	7e434bf3-■42d7-■4c1b-■b63b-■dcf52c7b18ff
INVESTIGATOR:	SOC Lead Analyst	**CLASSIFICATION:**	**CONFIDENTIAL // EVIDENTIARY DISCLOSURE**
EMAIL SHA-256:	`5e58c282c0ac76456bf139f43214■0e0dfc5338ca13502c9bea990d35■aa825c3b`	**HASH ALGORITHM:**	FIPS PUB 180-4 SHA-256 (Preserved Literal)

1. Executive Forensic Summary

DETERMINISTIC THREAT SCORE	ASSESSED SEVERITY	ML PHISHING PROBABILITY	ML CLASSIFICATION
48/100	**SUSPICIOUS**	**65.5%**	**PHISHING**

Email risk assessed as Suspicious (48/100) driven by 4 forensic signal(s): Possible PayPal lookalike domain (+18), Suspicious URL detected (+8), DNS Resolution Failure (+10).

2. Email Headers & Sender Alignment

From:	`security-■alert@■paypal-■update.■com`
To:	`victim@■company.■com`
Subject:	**Urgent Security Alert:■ Account Suspicious Access**
Date:	Tue,■ 08 Sep 2026 10:■30:■00 +■0000
Reply-To:	`None`
Return-Path:	`None`
Message-ID:	`&■lt;■evd-■test-■12345@■paypal-■update.■com&■gt;■`

From Domain:	`paypal-■update.■com`	**Reply-To Domain:**	N/■A **(Aligned)**
Return-Path Domain:	N/■A **(Aligned)**	**Alignment Verdict:**	**Sender domains aligned**

3. Email Authentication Findings (SPF / DKIM / DMARC)

PROTOCOL	OBSERVED VERDICT	TECHNICAL DETAILS & EVIDENCE
SPF (Sender Policy)	NONE	No Authentication-■Results or Received-■SPF header present.■
DKIM (Cryptographic)	NONE	No Authentication-■Results or DKIM-■Signature header present.■
DMARC (Domain Policy)	NONE	No Authentication-■Results header present.■

Notice: Observed authentication result from supplied headers (unverified by local mail server)

4. Deterministic Threat Score Breakdown

CATEGORY	IMPACT	FORENSIC FINDING & JUSTIFICATION
General	**+18 pts**	N/A

General	**+8 pts**	N/A
General	**+10 pts**	N/A
General	**+12 pts**	N/A

5. NLP Content Phishing Classification

Model Name:	TF-■IDF +■ Logistic Regression	**Classification:**	**PHISHING**
Phishing Probability:	**65.5%**	**Model Confidence:**	medium
Top Text Features:	`httpurl,■ account,■ immediately,■ security,■ urgent`	**Signal Role:**	Auxiliary ML signal (Non-overriding)

6. AI Analyst Assistant Synthesis

AI Analyst Assessment not requested or disabled for this analysis session.

7. Extracted Indicators of Compromise (IOCs)

Observed URLs & Suspicion Scoring:

EXTRACTED URL / TARGET HOST	SCORE	FORENSIC OBSERVATIONS & FLAGS
`http:■/■/■paypal-■update.■com/■login`	**30/100**	Potential brand impersonation:■ similar to paypal.■com (90%■ match),■ Suspicious security/■authentication keywords found:■ login,■ update

Observed Attachments & Cryptographic Checksums:

No file attachments detected in this message.

8. Network Transmission Path & Topology

Earliest Observable Sending IP:	**None identified**	**Reverse DNS / Host:**	Unknown

No sequential relay hops reconstructed from Received: headers.

Notice: Labeled strictly as observed infrastructure route. Does not represent physical attacker travel path.

9. Infrastructure & Domain Intelligence

No public IP intelligence enriched for this email.

10. Brand Impersonation & Lookalike Domains

SUSPECTED BRAND	SPOOFED DOMAIN	SIMILARITY	TECHNIQUES & ATTACK PROFILE
PayPal	`paypal-■update.■com`	**90.0%**	hyphenation,■ brand_■keyword,■ added_■affix

11. Prescriptive Recommendations & Containment Actions

[ ] 1. Block observed sender address and domain at the secure email gateway (SEG).
[ ] 2. Enforce domain-level DNS sinkholing for all identified suspicious/lookalike URLs.
[ ] 3. Revoke active user session tokens if credentials were submitted to target endpoints.
[ ] 4. Search tenant mailboxes for identical Message-ID, Subject, or attachment hashes to purge residual exposure.
[ ] 5. Submit extracted attachment hashes to Endpoint Detection and Response (EDR) blocklists.

12. Evidentiary Limitations & Chain-of-Custody Certification

Forensic Chain-of-Custody Certification:
This document is generated by MailTrace AI based on literal RFC-822 evidence headers. Public IP geolocation represents coarse regional estimates based on RIR allocations and does not establish street-level physical locations or the personal identity of the sender. All cryptographic hash sums (SHA-256) were computed over unaltered message byte streams and preserve complete digital non-repudiation chain of custody.

EVIDENTIARY DIGITAL SEAL:	`SHA256:5e58c282c0ac76456bf139f43214■0e0dfc5338ca13502c9bea990d35■aa825c3b`	**VERIFICATION STATUS:**	**CRYPTOGRAPHICALLY VERIFIED**